

```
jeniffan@apache2: ~/LAB1/Alice
jeniffan@apache2:~$ cd ../Bob~
bash: cd: ../Bob~: Aucun fichier ou dossier de ce nom
jeniffan@apache2:~$ mkdir LAB1
jeniffan@apache2:~$ cd LAB1
jeniffan@apache2:~/LAB1$ mkdir Alice
jeniffan@apache2:~/LAB1$ mkdir Bob
jeniffan@apache2:~/LAB1$ cd Alice
jeniffan@apache2:~/LAB1/Alice$ echo "Bonjour Bob, je suis Alice" > AliceDocument
jeniffan@apache2:~/LAB1/Alice$ openssl genrsa -out AliceKeyPair
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AliceKeyPair
jeniffan@apache2:~/LAB1/Alice$
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jeniffan@apache2:~/LAB1/Alice
jeniffan@apache2:~$ cd ../Bob~
bash: cd: ../Bob~: Aucun fichier ou dossier de ce nom
jeniffan@apache2:~$ mkdir LAB1
jeniffan@apache2:~$ cd LAB1
jeniffan@apache2:~/LAB1$ mkdir Alice
jeniffan@apache2:~/LAB1$ mkdir Bob
jeniffan@apache2:~/LAB1$ cd Alice
jeniffan@apache2:~/LAB1/Alice$ echo "Bonjour Bob, je suis Alice" > AliceDocument
jeniffan@apache2:~/LAB1/Alice$ openssl genrsa -out AliceKeyPair
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AliceKeyPair
jeniffan@apache2:~/LAB1/Alice$ openssl rsa -in AliceKeyPair -pubout -out AlicePublicKey
writing RSA key
jeniffan@apache2:~/LAB1/Alice$ mv AliceKeyPair AlicePrivateKey
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AlicePrivateKey AlicePublicKey
jeniffan@apache2:~/LAB1/Alice$
```

```
jeniffan@apache2: ~/LAB1/Bob
jeniffan@apache2:~$ mkdir LAB1
jeniffan@apache2:~$ cd LAB1
jeniffan@apache2:~/LAB1$ mkdir Alice
jeniffan@apache2:~/LAB1$ mkdir Bob
jeniffan@apache2:~/LAB1$ cd Alice
jeniffan@apache2:~/LAB1/Alice$ echo "Bonjour Bob, je suis Alice" > AliceDocument
jeniffan@apache2:~/LAB1/Alice$ openssl genrsa -out AliceKeyPair
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AliceKeyPair
jeniffan@apache2:~/LAB1/Alice$ openssl rsa -in AliceKeyPair -pubout -out AlicePublicKey
writing RSA key
jeniffan@apache2:~/LAB1/Alice$ mv AliceKeyPair AlicePrivateKey
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AlicePrivateKey AlicePublicKey
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ openssl genrsa -out BobKeyPair
jeniffan@apache2:~/LAB1/Bob$ openssl rsa -in BobKeyPair -pubout -out BobPublicKey
writing RSA key
jeniffan@apache2:~/LAB1/Bob$ mv BobKeyPair BobPrivateKey
jeniffan@apache2:~/LAB1/Bob$ ls
BobPrivateKey BobPublicKey
jeniffan@apache2:~/LAB1/Bob$
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jeniffan@apache2: ~/LAB1/Alice
jeniffan@apache2:~/LAB1$ cd Alice
jeniffan@apache2:~/LAB1/Alice$ echo "Bonjour Bob, je suis Alice" > AliceDocument
jeniffan@apache2:~/LAB1/Alice$ openssl genrsa -out AliceKeyPair
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument  AliceKeyPair
jeniffan@apache2:~/LAB1/Alice$ openssl rsa -in AliceKeyPair -pubout -out AlicePublicKey
writing RSA key
jeniffan@apache2:~/LAB1/Alice$ mv AliceKeyPair AlicePrivateKey
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument  AlicePrivateKey  AlicePublicKey
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ openssl genrsa -out BobKeyPair
jeniffan@apache2:~/LAB1/Bob$ openssl rsa -in BobKeyPair -pubout -out BobPublicKey
writing RSA key
jeniffan@apache2:~/LAB1/Bob$ mv BobKeyPair BobPrivateKey
jeniffan@apache2:~/LAB1/Bob$ ls
BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ cp BobPublicKey ../Alice/
jeniffan@apache2:~/LAB1/Bob$ cd ../Alice
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument  AlicePrivateKey  AlicePublicKey  BobPublicKey
jeniffan@apache2:~/LAB1/Alice$

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jeniffan@apache2: ~/LAB1/Alice
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ openssl genrsa -out BobKeyPair
jeniffan@apache2:~/LAB1/Bob$ openssl rsa -in BobKeyPair -pubout -out BobPublicKey
writing RSA key
jeniffan@apache2:~/LAB1/Bob$ mv BobKeyPair BobPrivateKey
jeniffan@apache2:~/LAB1/Bob$ ls
BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ cp BobPublicKey ../Alice/
jeniffan@apache2:~/LAB1/Bob$ cd ../Alice
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument  AlicePrivateKey  AlicePublicKey  BobPublicKey
jeniffan@apache2:~/LAB1/Alice$ openssl rsautl -encrypt -in AliceDocument -pubin -inkey BobPublicKey -out AliceDocumentEncrypted
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -encrypt -in AliceDocument -pubin -inkey BobPublicKey -out AliceDocumentEncrypted
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument      AlicePrivateKey  BobPublicKey
AliceDocumentEncrypted  AlicePublicKey
jeniffan@apache2:~/LAB1/Alice$ cat AliceDocumentEncrypted
00bH9~w0
0|0&00k000Bo0*0C0S01h00A      qJs0k200g00L002 5v00000#
y0v30nRtj000lzsZ0000Q0S00>4E00f
/q0^00of00l0y0000j0<0000y000:X00b0[0n0/0S0q000F-00000[|6H0 0<00I+000D<Q0000M0C0000rW-vm00
jeniffan@apache2:~/LAB1/Alice$

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jeniffan@apache2: ~/LAB1/Bob
jeniffan@apache2:~/LAB1/Bob$ mv BobKeyPair BobPrivateKey
jeniffan@apache2:~/LAB1/Bob$ ls
BobPrivateKey BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ cp BobPublicKey ../Alice/
jeniffan@apache2:~/LAB1/Bob$ cd ../Alice
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AlicePrivateKey AlicePublicKey BobPublicKey
jeniffan@apache2:~/LAB1/Alice$ openssl rsautl -encrypt -in AliceDocument -pubin -inkey BobPubli
cKey -out AliceDocumentEncrypted
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -encrypt -in AliceDocument -pubin -inkey BobPubl
icKey -out AliceDocumentEncrypted
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AlicePrivateKey BobPublicKey
AliceDocumentEncrypted AlicePublicKey
jeniffan@apache2:~/LAB1/Alice$ cat AliceDocumentEncrypted
00bH9~w0
0|0&00k000Bo0*0C0S01h00A qJs0k200g00L002 0v00000#
y0v30nRtj000ljsZ0000Q0S00>4E00f
/q0^00of00l0y0000j0<0000y000:X00b0[0n0/0&0q000F-00000[6H0
000I+000D<Q00000M00C0000rW-vm00
jeniffan@apache2:~/LAB1/Alice$ cp AliceDocumentEncrypted ../Bob/
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentEncrypted BobPrivateKey BobPublicKey
jeniffan@apache2:~/LAB1/Bob$
```

```
jeniffan@apache2: ~/LAB1/Bob
AliceDocument AlicePrivateKey AlicePublicKey BobPublicKey
jeniffan@apache2:~/LAB1/Alice$ openssl rsautl -encrypt -in AliceDocument -pubin -inkey BobPublic
Key -out AliceDocumentEncrypted
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -encrypt -in AliceDocument -pubin -inkey BobPubl
icKey -out AliceDocumentEncrypted
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument          AlicePrivateKey  BobPublicKey
AliceDocumentEncrypted AlicePublicKey
jeniffan@apache2:~/LAB1/Alice$ cat AliceDocumentEncrypted
00bH9~w0
0|0&00k000Bo0*0C0S01h00A          qJs0k200g00L002 5v00000#
y0v30nRtj000ljzS0000Q0S00>4E00f
/q0^00of00l0y0000j0<0000y000:X00b0[0n0/0&0q000F-00000[|6H0 0_00I+000D<Q00000M0C0000rW-vm00
jeniffan@apache2:~/LAB1/Alice$ cp AliceDocumentEncrypted ../Bob/
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentEncrypted BobPrivateKey BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ openssl pkeyutl -decrypt -in AliceDocumentEncrypted -inkey BobPriv
ateKey -out AliceDocumentDecrypted
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentDecrypted AliceDocumentEncrypted BobPrivateKey BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ cat AliceDocumentDecrypted
Bonjour Bob, je suis Alice
jeniffan@apache2:~/LAB1/Bob$
```

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jeniffan@apache2: ~/LAB1/Alice
jeniffan@apache2:~/LAB1/Alice$ openssl rsautl -encrypt -in AliceDocument -pubin -inkey BobPublic
Key -out AliceDocumentEncrypted
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -encrypt -in AliceDocument -pubin -inkey BobPubl
icKey -out AliceDocumentEncrypted
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument      AlicePrivateKey  BobPublicKey
AliceDocumentEncrypted  AlicePublicKey
jeniffan@apache2:~/LAB1/Alice$ cat AliceDocumentEncrypted
00bH9~W0
0|0&00k000Bo0*0C0S01h00A      qJs0k200g00L002 5v00000#
y0v30nRtj0001jsZ0000Q0S00>4E00f
/q0^00of00l0y0000j0<0000y000:X00b0[0n0/0&0q000F-00000[|6H0 0500I+000D<Q0000M0C0000rW-vm00
jeniffan@apache2:~/LAB1/Alice$ cp AliceDocumentEncrypted ../Bob/
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentEncrypted  BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ openssl pkeyutl -decrypt -in AliceDocumentEncrypted -inkey BobPriv
ateKey -out AliceDocumentDecrypted
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentEncrypted  AliceDocumentDecrypted  BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ cat AliceDocumentDecrypted
Bonjour Bob, je suis Alice
jeniffan@apache2:~/LAB1/Bob$ cd ../Alice
jeniffan@apache2:~/LAB1/Alice$

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jeniffan@apache2: ~/LAB1/Alice
AliceDocument      AlicePrivateKey  BobPublicKey
AliceDocumentEncrypted  AlicePublicKey
jeniffan@apache2:~/LAB1/Alice$ cat AliceDocumentEncrypted
00bH9~W0
0|0&00k000Bo0*0C0S01h00A      qJs0k200g00L002 5v00000#
y0v30nRtj0001jsZ0000Q0S00>4E00f
/q0^00of00l0y0000j0<0000y000:X00b0[0n0/0&0q000F-00000[|6H0 0500I+000D<Q0000M0C0000rW-vm00
jeniffan@apache2:~/LAB1/Alice$ cp AliceDocumentEncrypted ../Bob/
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentEncrypted  BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ openssl pkeyutl -decrypt -in AliceDocumentEncrypted -inkey BobPriv
ateKey -out AliceDocumentDecrypted
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentDecrypted  AliceDocumentEncrypted  BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ cat AliceDocumentDecrypted
Bonjour Bob, je suis Alice
jeniffan@apache2:~/LAB1/Bob$ cd ../Alice
jeniffan@apache2:~/LAB1/Alice$ openssl rand -out LargeFile -base64 $((2*30*3/4))
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -encrypt -in LargeFile -pubin -inkey BobPublicKe
y -out LargeFileEncrypted
Public Key operation error
20501084FFFF0000:error:0200006E:rsa routines:ossl_rsa_padding_add_PKCS1_type_2_ex:data too larg
e for key size:../crypto/rsa/rsa_pk1.c:132:
jeniffan@apache2:~/LAB1/Alice$

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jeniffan@apache2: ~/LAB1/Alice
|0&00k000Bo0*0C0S01h00A      qJs0k200g00L002  0v00000#
y0v30nRtj000ljsZ0000Q0S00>4E00f
/q0^00of00l0y0000j0<0000y000:X00b0[0n0/0&0q000F-00000[|6H0      000I+000D<Q0000M0C0000rW-vm00
jeniffan@apache2:~/LAB1/Alice$ cp AliceDocumentEncrypted ../Bob/
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentEncrypted  BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ openssl pkeyutl -decrypt -in AliceDocumentEncrypted -inkey BobPriv
ateKey -out AliceDocumentDecrypted
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentDecrypted  AliceDocumentEncrypted  BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ cat AliceDocumentDecrypted
Bonjour Bob, je suis Alice
jeniffan@apache2:~/LAB1/Bob$ cd ../Alice
jeniffan@apache2:~/LAB1/Alice$ openssl rand -out LargeFile -base64 $((2**30*3/4))
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -encrypt -in LargeFile -pubin -inkey BobPublicKe
y -out LargeFileEncrypted
Public Key operation error
20501084FFFF0000:error:0200006E:rsa routines:ossl_rsa_padding_add_PKCS1_type_2_ex:data too larg
e for key size:../crypto/rsa/rsa_pk1.c:132:
jeniffan@apache2:~/LAB1/Alice$ echo "Ceci est une donnee d authentification" > AuthData
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument      AlicePrivateKey  AuthData      LargeFile
AliceDocumentEncrypted  AlicePublicKey  BobPublicKey  LargeFileEncrypted
jeniffan@apache2:~/LAB1/Alice$

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jeniffan@apache2: ~/LAB1/Alice
y0v30nRtj000ljsZ0000Q0S00>4E00f
/q0^00of00l0y0000j0<0000y000:X00b0[0n0/0&0q000F-00000[|6H0      000I+000D<Q0000M0C0000rW-vm00
jeniffan@apache2:~/LAB1/Alice$ cp AliceDocumentEncrypted ../Bob/
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentEncrypted  BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ openssl pkeyutl -decrypt -in AliceDocumentEncrypted -inkey BobPriv
ateKey -out AliceDocumentDecrypted
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentDecrypted  AliceDocumentEncrypted  BobPrivateKey  BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ cat AliceDocumentDecrypted
Bonjour Bob, je suis Alice
jeniffan@apache2:~/LAB1/Bob$ cd ../Alice
jeniffan@apache2:~/LAB1/Alice$ openssl rand -out LargeFile -base64 $((2**30*3/4))
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -encrypt -in LargeFile -pubin -inkey BobPublicKe
y -out LargeFileEncrypted
Public Key operation error
20501084FFFF0000:error:0200006E:rsa routines:ossl_rsa_padding_add_PKCS1_type_2_ex:data too larg
e for key size:../crypto/rsa/rsa_pk1.c:132:
jeniffan@apache2:~/LAB1/Alice$ echo "Ceci est une donnee d authentification" > AuthData
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument      AlicePrivateKey  AuthData      LargeFile
AliceDocumentEncrypted  AlicePublicKey  BobPublicKey  LargeFileEncrypted
jeniffan@apache2:~/LAB1/Alice$ cp AlicePublicKey ../Bob/
jeniffan@apache2:~/LAB1/Alice$

```

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jeniffan@apache2: ~/LAB1/Alice
jeniffan@apache2:~/LAB1/Bob$ openssl pkeyutl -decrypt -in AliceDocumentEncrypted -inkey BobPrivateKey -out AliceDocumentDecrypted
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentDecrypted AliceDocumentEncrypted BobPrivateKey BobPublicKey
jeniffan@apache2:~/LAB1/Bob$ cat AliceDocumentDecrypted
Bonjour Bob, je suis Alice
jeniffan@apache2:~/LAB1/Bob$ cd ../Alice
jeniffan@apache2:~/LAB1/Alice$ openssl rand -out LargeFile -base64 $((2**30*3/4))
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -encrypt -in LargeFile -pubin -inkey BobPublicKey -out LargeFileEncrypted
Public Key operation error
20501084FFFF0000:error:0200006E:rsa routines:ossl_rsa_padding_add_PKCS1_type_2_ex:data too large for key size:../crypto/rsa/rsa_pk1.c:132:
jeniffan@apache2:~/LAB1/Alice$ echo "Ceci est une donnee d authentication" > AuthData
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AlicePrivateKey AuthData LargeFile
AliceDocumentEncrypted AlicePublicKey BobPublicKey LargeFileEncrypted
jeniffan@apache2:~/LAB1/Alice$ cp AlicePublicKey ../Bob/
jeniffan@apache2:~/LAB1/Alice$ openssl dgst -sha256 -out HashAuthData AuthData
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AlicePrivateKey AuthData HashAuthData LargeFileEncrypted
AliceDocumentEncrypted AlicePublicKey BobPublicKey LargeFile
jeniffan@apache2:~/LAB1/Alice$ cat HashAuthData
SHA2-256(AuthData)= 4c18416007b0aed2a28b275ab9f0d7f1650a33b9155b1e775509638c37fa64dd
jeniffan@apache2:~/LAB1/Alice$

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jeniffan@apache2: ~/LAB1/Alice
y -out LargeFileEncrypted
Public Key operation error
20501084FFFF0000:error:0200006E:rsa routines:ossl_rsa_padding_add_PKCS1_type_2_ex:data too large for key size:../crypto/rsa/rsa_pk1.c:132:
jeniffan@apache2:~/LAB1/Alice$ echo "Ceci est une donnee d authentication" > AuthData
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AlicePrivateKey AuthData LargeFile
AliceDocumentEncrypted AlicePublicKey BobPublicKey LargeFileEncrypted
jeniffan@apache2:~/LAB1/Alice$ cp AlicePublicKey ../Bob/
jeniffan@apache2:~/LAB1/Alice$ openssl dgst -sha256 -out HashAuthData AuthData
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AlicePrivateKey AuthData HashAuthData LargeFileEncrypted
AliceDocumentEncrypted AlicePublicKey BobPublicKey LargeFile
jeniffan@apache2:~/LAB1/Alice$ cat HashAuthData
SHA2-256(AuthData)= 4c18416007b0aed2a28b275ab9f0d7f1650a33b9155b1e775509638c37fa64dd
jeniffan@apache2:~/LAB1/Alice$ openssl rsautl -sign -in HashAuthData -inkey AlicePrivateKey -out AliceSignature
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -sign -in HashAuthData -inkey AlicePrivateKey -out AliceSignature
Error: The non-raw input data length 85 is too long - max supported hashed size is 64
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument AlicePrivateKey AliceSignature BobPublicKey LargeFile
AliceDocumentEncrypted AlicePublicKey AuthData HashAuthData LargeFileEncrypted
jeniffan@apache2:~/LAB1/Alice$

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jeniffan@apache2: ~/LAB1/Bob
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument      AlicePrivateKey  AuthData      LargeFile
AliceDocumentEncrypted AlicePublicKey  BobPublicKey  LargeFileEncrypted
jeniffan@apache2:~/LAB1/Alice$ cp AlicePublicKey ../Bob/
jeniffan@apache2:~/LAB1/Alice$ openssl dgst -sha256 -out HashAuthData AuthData
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument      AlicePrivateKey  AuthData      HashAuthData  LargeFileEncrypted
AliceDocumentEncrypted AlicePublicKey  BobPublicKey  LargeFile
jeniffan@apache2:~/LAB1/Alice$ cat HashAuthData
SHA2-256(AuthData)= 4c18416007b0aed2a28b275ab9f0d7f1650a33b9155b1e775509638c37fa64dd
jeniffan@apache2:~/LAB1/Alice$ openssl rsautl -sign -in HashAuthData -inkey AlicePrivateKey -out AliceSignature
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
jeniffan@apache2:~/LAB1/Alice$ openssl pkeyutl -sign -in HashAuthData -inkey AlicePrivateKey -out AliceSignature
Error: The non-raw input data length 85 is too long - max supported hashed size is 64
jeniffan@apache2:~/LAB1/Alice$ ls
AliceDocument      AlicePrivateKey  AliceSignature  BobPublicKey  LargeFile
AliceDocumentEncrypted AlicePublicKey  AuthData      HashAuthData  LargeFileEncrypted
jeniffan@apache2:~/LAB1/Alice$ cp AliceSignature AuthData ../Bob/
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ ls
AliceDocumentDecrypted AlicePublicKey  AuthData      BobPublicKey
AliceDocumentEncrypted AliceSignature  BobPrivateKey
jeniffan@apache2:~/LAB1/Bob$

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jeniffan@apache2: ~/LAB1/Bob
20C02DA0FFFF0000:error:0200006A:rsa routines:RSA_padding_check_PKCS1_type_1:block type is not 0
1:../crypto/rsa/rsa_pk1.c:85:
20C02DA0FFFF0000:error:02000072:rsa routines:rsa_ossl_public_decrypt:padding check failed:../crypto/rsa/rsa_ossl.c:788:
20C02DA0FFFF0000:error:1C880004:Provider routines:rsa_verify_recover:RSA lib:../providers/implementations/signature/rsa_sig.c:1000:
20C02DA0FFFF0000:error:030000EA:digital envelope routines:EVP_PKEY_verify_recover:provider signature failure:../crypto/evp/signature.c:1236:RSA verify_recover:
jeniffan@apache2:~/LAB1/Bob$ openssl dgst -sha256 -out HashBob AuthData
jeniffan@apache2:~/LAB1/Bob$ diff HashBob HashAuthData
1d0
< SHA2-256(AuthData)= 4c18416007b0aed2a28b275ab9f0d7f1650a33b9155b1e775509638c37fa64dd
jeniffan@apache2:~/LAB1/Bob$ cd ../Alice
jeniffan@apache2:~/LAB1/Alice$ openssl dgst -sha256 -out HashAuthData AuthData
jeniffan@apache2:~/LAB1/Alice$ openssl rsautl -sign -in HashAuthData -inkey AlicePrivateKey -out AliceSignature
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
jeniffan@apache2:~/LAB1/Alice$ cp AliceSignature ../Bob/
jeniffan@apache2:~/LAB1/Alice$ cd ../Bob
jeniffan@apache2:~/LAB1/Bob$ openssl rsautl -verify -in AliceSignature -pubin -inkey AlicePublicKey -out HashAuthData
The command rsautl was deprecated in version 3.0. Use 'pkeyutl' instead.
jeniffan@apache2:~/LAB1/Bob$ openssl dgst -sha256 -out HashBob AuthData
jeniffan@apache2:~/LAB1/Bob$ diff HashBob HashAuthData
jeniffan@apache2:~/LAB1/Bob$

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